

Equivalence of Progress Related Non-Interference Definitions

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Definitions (D)

1. Definitions of Notable Terms:

- (a) let $\mathbb{T}$ be the set of all traces, where each trace is a possibly infinite sequence of events
- (b) let **Store** be the set of all memory stores
- (c) let $\mathbb{T}_{\text{fin}}$ be the set of all finite traces
- (d) let $\mathbb{E}$ denote the set of all possible events
- (e) let $\mathcal{D}$ denote the low (or public) context/observer

2. Hyperproperty Definition of PsNi:

For an arbitrary set of traces, initial memory store pairs $T \subseteq \mathbf{Store} \times \mathbb{T}$:

$$T \in \text{HPsNi}_{\mathcal{D}} \iff \forall (m_1, st_1), (m_2, st_2) \in T. m_1 \approx_{\mathcal{D}} m_2 \implies \forall p_1 \in \mathbb{T}_{\text{fin}} \leq st_1. \exists p_2 \in \mathbb{T}_{\text{fin}} \leq st_2. p_1 \approx_{\mathcal{D}} p_2$$

3. Definition of Program behavior:

For an arbitrary program c :

$$\text{Behav}(c) \subseteq \mathbf{Store} \times \mathbb{T} = \{(m, st) \in \mathbf{Store} \times \mathbb{T} \mid c \rightsquigarrow (m, st)\}$$

4. Definition of (Deterministic) Program Trace Production:

For a program c and arbitrary store, trace pair $(m, st) \in \mathbf{Store} \times \mathbb{T}$:

$$c \rightsquigarrow (m, st) \iff \forall p \in \mathbb{T}_{\text{fin}}. (p \leq st \iff \langle c, m \rangle \rightarrow_p^*)$$

5. Definition of Program, Memory Production of Finite Trace:

For an arbitrary program c , memory store $m \in \mathbf{Store}$, and finite trace $p \in \mathbb{T}_{\text{fin}}$

$$\langle c, m \rangle \rightarrow_p^* \iff \exists c', m' \in \mathbf{Store}. \langle c, m \rangle \rightarrow_p^* \langle c', m' \rangle$$

6. Definition of Attacker Knowledge:

For a program c , an initial memory $m \in \mathbf{Store}$, and finite trace $p \in \mathbb{T}_{\text{fin}}$ where $\langle c, m \rangle \rightarrow_p^*$, the attacker knowledge of c with m at p is defined as:

$$k(c, m, p, \mathcal{D}) = \{m' \mid m' \approx_{\mathcal{D}} m \wedge \exists p' \in \mathbb{T}_{\text{fin}}. (\langle c, m' \rangle \rightarrow_{p'}^* \wedge p \approx_{\mathcal{D}} p')\}$$

7. Knowledge definition of PsNi:

for an arbitrary program c :

$$c \in \text{KPsNi}_{\mathcal{D}} \iff \forall p \in \mathbb{T}_{\text{fin}}, m \in \mathbf{Store}, \alpha \in \mathbb{E}. \langle c, m \rangle \rightarrow_{p \cdot \alpha}^* \wedge \neg \text{Sil}_{\mathcal{D}}(\alpha) \implies k(c, m, p, \mathcal{D}) = k(c, m, p \cdot \alpha, \mathcal{D})$$

8. Hyperproperty definition of PiNi:

For an arbitrary set of traces, initial memory store pairs $T \subseteq \mathbf{Store} \times \mathbb{T}$:

$$T \in \text{HPiNi}_{\mathcal{D}} \iff \forall (m_1, st_1), (m_2, st_2) \in T. m_1 \approx_{\mathcal{D}} m_2 \implies \forall p_1 \in \mathbb{T}_{\text{fin}} \leq st_1, p_2 \in \mathbb{T}_{\text{fin}} \leq st_2. (p_1 \leq_{\mathcal{D}} p_2 \vee p_2 \leq_{\mathcal{D}} p_1)$$

9. Definition of low prefix:

For two finite traces $p_1, p_2 \in \mathbb{T}_{\text{fin}}$, the low prefix relation is defined as

$$p_1 \leq_{\mathcal{D}} p_2 \iff \exists p'_2 \leq p_2. p_1 \approx_{\mathcal{D}} p'_2$$

10. **Definition of Progress Knowledge:**

For a program c , an initial memory $m \in \mathbf{Store}$, and finite trace $p \in \mathbb{T}_{\text{fin}}$ where $\langle c, m \rangle \rightarrow_p^*$, the Progress knowledge of c with m at p is defined as:

$$k_{\rightarrow}(c, m, p, \mathcal{D}) = \{m' \mid m' \approx_{\mathcal{D}} m \wedge \exists p' \in \mathbb{T}_{\text{fin}}, \alpha' \in \mathbb{E}. (\langle c, m' \rangle \rightarrow_{p'}^* \wedge \neg \text{Sil}_{\mathcal{D}}(\alpha') \wedge p' \approx_{\mathcal{D}} p \cdot \alpha')\}$$

11. **Knowledge based definition of PiNi:**

For an arbitrary program c :

$$c \in \text{KPiNi}_{\mathcal{D}} \stackrel{\Delta}{\iff} \forall p \in \mathbb{T}_{\text{fin}}, m \in \mathbf{Store}, \alpha \in \mathbb{E}. \langle c, m \rangle \rightarrow_{p \cdot \alpha}^* \wedge \neg \text{Sil}_{\mathcal{D}}(\alpha) \implies k_{\rightarrow}(c, m, p, \mathcal{D}) = k(c, m, p \cdot \alpha, \mathcal{D})$$

12. **Definition of Non-Silent Trace Progress:**

For an arbitrary program c with trace t and memory m s.t. $c \rightsquigarrow (m, t)$ and $p \leq t$. The assertion that t has an additional non-silent event beyond p is defined as:

$$\text{Prog}_{\mathcal{D}}(p, t) \stackrel{\Delta}{\iff} \exists p' \leq t. p <_{\mathcal{D}} p'$$

13. **Hyperproperty definition of Lfp:**

For an arbitrary set of traces, initial memory store pairs $T \subseteq \mathbf{Store} \times \mathbb{T}$:

$$T \in \text{HLfp}_{\mathcal{D}} \stackrel{\Delta}{\iff} \forall (m_1, st_1), (m_2, st_2) \in T. m_1 \approx_{\mathcal{D}} m_2 \implies \forall p_1 \leq st_1, p_2 \leq st_2 \in \mathbb{T}_{\text{fin}}. p_1 <_{\mathcal{D}} p_2 \implies \text{Prog}_{\mathcal{D}}(p_1, st_1)$$

14. **Knowledge based definition of Lfp:**

For an arbitrary program c :

$$c \in \text{KLfp}_{\mathcal{D}} \stackrel{\Delta}{\iff} \forall p \in \mathbb{T}_{\text{fin}}, m \in \mathbf{Store}, \alpha \in \mathbb{E}. \neg \text{Sil}(\alpha) \wedge \langle c, m \rangle \rightarrow_{p \cdot \alpha}^* \implies k(c, m, p, \mathcal{D}) = k_{\rightarrow}(c, m, p, \mathcal{D})$$

Notable Facts (NF)

1. **Determinism:**

If a program c is deterministic, then it follows that:

$$\langle c, m \rangle \xrightarrow{\alpha_1} \langle c_1, m_1 \rangle \wedge \langle c, m \rangle \xrightarrow{\alpha_2} \langle c_2, m_2 \rangle \implies c_1 = c_2 \wedge m_1 = m_2 \wedge \alpha_1 = \alpha_2$$

2. **Trace Prefix Maximization:**

Given program c as well as arbitrary $p \in \mathbb{T}_{\text{fin}}, m \in \mathbf{Store}$.

$$\langle c, m \rangle \rightarrow_p^* \implies \exists st \in \mathbb{T}. p \leq st \wedge c \rightsquigarrow (m, st)$$

3. **Alternative Formulation of Indistinguishable Traces:**

using the erasure function $\lfloor \cdot \rfloor_{\mathcal{D}}$: $p_1 \approx_{\mathcal{D}} p_2$ and $p_1 \leq_{\mathcal{D}} p_2$ can be obtained as follows:

$$p_1 <_{\mathcal{D}} p_2 \iff \lfloor p_1 \rfloor_{\mathcal{D}} < \lfloor p_2 \rfloor_{\mathcal{D}}$$

$$p_1 \approx_{\mathcal{D}} p_2 \iff \lfloor p_1 \rfloor_{\mathcal{D}} = \lfloor p_2 \rfloor_{\mathcal{D}}$$

PsNi Proofs (PS)

1. Forward Proof (from Hyperproperty PsNi to Knowledge PsNi)

Given that for an arbitrary program c , $\text{Behav}(c) \in \text{HPsNi}_{\mathcal{D}}$, show that $c \in \text{KPsNi}_{\mathcal{D}}$.

Lemma 1.1: if $\text{Behav}(c) \in \text{HPsNi}_{\mathcal{D}}$, then for any $(m_1, st_1), (m_2, st_2) \in \text{Behav}(c)$:

$$m_1 \approx_{\mathcal{D}} m_2 \iff m_1 \approx_{\mathcal{D}} m_2 \wedge \forall p_1 \in \mathbb{T}_{\text{fin}} \leq st_1. \exists p_2 \in \mathbb{T}_{\text{fin}} \leq st_2. (\langle c, m_2 \rangle \rightarrow_{p_2}^* \wedge p_2 \approx_{\mathcal{D}} p_1)$$

Proof: From the premise, assume that $\text{Behav}(c) \in \text{HPsNi}_{\mathcal{D}}$. Take $(m_1, st_1), (m_2, st_2) \in \text{Behav}(c)$ for which $m_1 \approx_{\mathcal{D}} m_2$.

(a) Consider first the claim:

$$m_1 \approx_{\mathcal{D}} m_2 \implies m_1 \approx_{\mathcal{D}} m_2 \wedge \forall p_1 \in \mathbb{T}_{\text{fin}} \leq st_1. \exists p_2 \in \mathbb{T}_{\text{fin}} \leq st_2. (\langle c, m_2 \rangle \rightarrow_{p_2}^* \wedge p_2 \approx_{\mathcal{D}} p_1)$$

Let arbitrary $p_1 \in \mathbb{T}_{\text{fin}} \leq st_1$. Since $m_1 \approx_{\mathcal{D}} m_2$ by assumption, applying definition (D.2) knowing $\text{Behav}(c) \in \text{HPsNi}_{\mathcal{D}}$ implies that:

$$\exists p_2 \leq st_2 \in \mathbb{T}_{\text{fin}}. p_1 \approx_{\mathcal{D}} p_2$$

Then applying definition (D.4) given that $p_2 \leq st_2$ shows that $\langle c, m_2 \rangle \rightarrow_{p_2}^*$ as well.

Since p_1 was an arbitrarily chosen prefix of st_1 , the two results above suffice to prove the claim.

(b) The converse:

$$m_1 \approx_{\mathcal{D}} m_2 \iff m_1 \approx_{\mathcal{D}} m_2 \wedge \forall p_1 \in \mathbb{T}_{\text{fin}} \leq st_1. \exists p_2 \in \mathbb{T}_{\text{fin}} \leq st_2. (\langle c, m_2 \rangle \rightarrow_{p_2}^* \wedge p_2 \approx_{\mathcal{D}} p_1)$$

follows immediately from the rules of logical conjunction

Lemma 1.2: if $\text{Behav}(c) \in \text{HPsNi}_{\mathcal{D}}$ then $\forall (m, st) \in \text{Behav}(c). k(c, m, p, \mathcal{D}) = \{m' \mid m' \approx_{\mathcal{D}} m\}$

Since (m_1, st_1) and (m_2, st_2) were arbitrarily chosen in lemma 1.1, Applying lemma 1.1 with definition (D.6) on any $(m, st) \in \text{Behav}(c)$ with any prefix $p \in \mathbb{T}_{\text{fin}} \leq st$ shows that:

$$\begin{aligned} k(c, m, p, \mathcal{D}) &= \{m' \mid m' \approx_{\mathcal{D}} m \wedge \exists p' \in \mathbb{T}_{\text{fin}}. (\langle c, m' \rangle \rightarrow_{p'}^* \wedge p \approx_{\mathcal{D}} p')\} \\ &= \{m' \mid m' \approx_{\mathcal{D}} m \wedge \top\} = \{m' \mid m' \approx_{\mathcal{D}} m\} \end{aligned}$$

The claim $c \in \text{KPsNi}_{\mathcal{D}}$ follows immediately from lemma 1.2, as for any arbitrary choice of $m \in \text{Store}$, $p \in \mathbb{T}_{\text{fin}}$, and $\alpha \in \mathbb{E}$: if $\langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$ then:

$$k(c, m, p, \mathcal{D}) = \{m' \mid m' \approx_{\mathcal{D}} m\} = k(c, m, p \cdot \alpha, \mathcal{D})$$

2. Backward Proof (from Knowledge PsNi to Hyperproperty PsNi)

Given that for an arbitrary program c where $c \in \text{KPsNi}_{\mathcal{D}}$, show that $\text{Behav}(c) \in \text{HPsNi}_{\mathcal{D}}$.

Lemma 2.1: if $c \in \text{KPsNi}_{\mathcal{D}}$, then $\forall p \in \mathbb{T}_{\text{fin}}, m \in \text{Store}. \langle c, m \rangle \rightarrow_p^* \implies k(c, m, p, \mathcal{D}) = \{m' \mid m' \approx_{\mathcal{D}} m\}$

Proof: Given an existing initial memory m , Lemma 2.1 can be proved by induction on finite traces p .

Base Case: for the base case, consider the empty trace $p = \epsilon$. Since for all initial memory stores $m', \langle c, m' \rangle \rightarrow_{\epsilon}^* \langle c, m' \rangle$ and $\epsilon \approx_{\mathcal{D}} \epsilon$ trivially, it then follows that:

$$\begin{aligned} k(c, m, \epsilon, \mathcal{D}) &= \{m' \mid m' \approx_{\mathcal{D}} m \wedge \exists p' \in \mathbb{T}_{\text{fin}}. (\langle c, m' \rangle \rightarrow_{p'}^* \wedge \epsilon \approx_{\mathcal{D}} p')\} \\ &= \{m' \mid m' \approx_{\mathcal{D}} m\} \end{aligned}$$

inductive Case: As the inductive hypothesis, for $p \in \mathbb{T}_{\text{fin}}: \langle c, m \rangle \rightarrow_p^* \implies k(c, m, p, \mathcal{D}) = \{m' \mid m' \approx_{\mathcal{D}} m\}$.

Then for an event $\alpha \in \mathbb{E}$ where $\langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$, lemma 2.1 can be extended to trace $p \cdot \alpha$ after applying definition (D.7) and then applying the inductive hypothesis (as $\langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$ implies $\langle c, m \rangle \rightarrow_p^*$) gives:

$$k(c, m, p \cdot \alpha, \mathcal{D}) = k(c, m, p, \mathcal{D}) = \{m' \mid m' \approx_{\mathcal{D}} m\}$$

Let $(m_1, st_1), (m_2, st_2) \in \text{Behav}(c)$ be arbitrary memory store, trace pairs where $c \rightsquigarrow (m_1, st_1)$ and $c \rightsquigarrow (m_2, st_2)$, subject to the constraint that $m_1 \approx_{\mathcal{D}} m_2$.

Let $p_1 \in \mathbb{T}_{\text{fin}} \leq st_1$, from (D.4), it follows that $\langle c, m_1 \rangle \rightarrow_{p_1}^*$.

From lemma 2.1, $k(c, m_1, p_1, \mathcal{D}) = \{m \mid m \approx_{\mathcal{D}} m_1\}$.

As $m_1 \approx_{\mathcal{D}} m_2$ from the premise, it follows that $m_2 \in k(c, m_1, p_1, \mathcal{D})$.

Then from (D.6), $\exists p \in \mathbb{T}_{\text{fin}}. \langle c, m_2 \rangle \rightarrow_p^* \wedge p \approx_{\mathcal{D}} p_1$, after which applying (D.4) with $c \rightsquigarrow (m_2, st_2)$ shows that $p \leq st_2$.

Using p as the witness then fulfills the required property that:

$$\exists p_2 \in \mathbb{T}_{\text{fin}} \leq st_2. p_1 \approx_{\mathcal{D}} p_2$$

Thus, since $(m_1, st_1), (m_2, st_2)$, and p_1 were arbitrarily chosen, the above statement suffices to show:

$$\forall (m_1, st_1), (m_2, st_2) \in \text{Behav}(c). (m_1 \approx_{\mathcal{D}} m_2 \implies \forall p_1 \in \mathbb{T}_{\text{fin}} \leq st_1. \exists p_2 \in \mathbb{T}_{\text{fin}} \leq st_2. p_1 \approx_{\mathcal{D}} p_2)$$

or that $\text{Behav}(c) \in \text{HPsNi}_{\mathcal{D}}$ by (D.2), as desired.

PiNi Proofs (PI)

1. Forward Proof – (Alternate Formulation) (from Hyperproperty PiNi to Knowledge PiNi)

Given that for an arbitrary program c , $\text{Behav}(c) \in \text{HPiNi}_{\mathcal{D}}$, show that $c \in \text{KPiNi}_{\mathcal{D}}$.

Lemma 3.1: if $\text{Behav}(c) \in \text{HPiNi}_{\mathcal{D}}$, then for any $m, m' \in \text{Store}$, $p \in \mathbb{T}_{\text{fin}}$, and $\alpha \in \mathbb{E}$ where $m \approx_{\mathcal{D}} m'$, $\neg \text{Sil}_{\mathcal{D}}(\alpha)$, and $\langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$:

$$\exists p', \alpha'. \langle c, m' \rangle \rightarrow_{p'}^* \wedge \neg \text{Sil}_{\mathcal{D}}(\alpha') \wedge p' \approx_{\mathcal{D}} p \cdot \alpha' \implies \exists p^*. \langle c, m' \rangle \rightarrow_{p^*}^* \wedge p \cdot \alpha \approx_{\mathcal{D}} p^*$$

Proof: Consider arbitrary $m \in \text{Store}$, $p \in \mathbb{T}_{\text{fin}}$, $\alpha \in \mathbb{E}$ such that $\langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$ (from premises).

From (NF.2), it follows that $\exists st \in \mathbb{T}$. $p \cdot \alpha \leq st$ and $c \rightsquigarrow (m, st)$.

Additionally, consider arbitrary $m' \in \text{Store}$ where $m \approx_{\mathcal{D}} m'$ and a particular $p' \in \mathbb{T}_{\text{fin}}$, $\alpha' \in \mathbb{E}$ such that $p \approx_{\mathcal{D}} p'$, $\langle c', m' \rangle \rightarrow_{p'}^*$, and $\neg \text{Sil}_{\mathcal{D}}(\alpha')$.

Also from (NF.2), it follows that $\exists st' \in \mathbb{T}$. $p' \leq st'$ and $c \rightsquigarrow (m', st')$.

Since $\text{Behav}(c) \in \text{HPiNi}_{\mathcal{D}}$ and $m \approx_{\mathcal{D}} m'$, it follows that:

$$\forall p_a \leq st, p_b \leq st'. p_a \leq_{\mathcal{D}} p_b \vee p_b \leq_{\mathcal{D}} p_a$$

then taking $p_a = p \cdot \alpha$ and $p_b = p'$ yields that $\alpha = \alpha'$ from case analysis and using (NF.3) with list properties:

(case 1)	(case 2)
$p \cdot \alpha \leq_{\mathcal{D}} p'$	$p' \leq_{\mathcal{D}} p \cdot \alpha$
$p \cdot \alpha \leq_{\mathcal{D}} p \cdot \alpha'$	$p \cdot \alpha' \leq_{\mathcal{D}} p \cdot \alpha$
$\alpha \leq_{\mathcal{D}} \alpha'$	$\alpha' \leq_{\mathcal{D}} \alpha$
$\alpha = \alpha'$	$\alpha' = \alpha$

After which the conclusion follows immediately, as $p' \approx_{\mathcal{D}} p \cdot \alpha' = p \cdot \alpha$, allowing p' to serve as witness p^* .

The claim $c \in \text{KPiNi}_{\mathcal{D}}$ then follows straightforwardly from applying lemma 3.1 with the progress and attacker knowledge definitions. For any arbitrary choice of $m \in \text{Store}$, $p \in \mathbb{T}_{\text{fin}}$, and $\alpha \in \mathbb{E}$: if $\langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$ and $\neg \text{Sil}_{\mathcal{D}}(\alpha)$ then:

$$\begin{aligned} k_{\rightarrow}(c, m, p, \mathcal{D}) &= \{m' \mid m' \approx_{\mathcal{D}} m \wedge \exists p' \in \mathbb{T}_{\text{fin}}, \alpha' \in \mathbb{E}. (\langle c, m' \rangle \rightarrow_{p' \cdot \alpha'}^* \wedge \neg \text{Sil}_{\mathcal{D}}(\alpha') \wedge p \approx_{\mathcal{D}} p')\} \\ &= \{m' \mid m' \approx_{\mathcal{D}} m \wedge \exists p^* \in \mathbb{T}_{\text{fin}}. (\langle c, m' \rangle \rightarrow_{p^*}^* \wedge p \cdot \alpha \approx_{\mathcal{D}} p^*)\} \\ &= k(c, m, p \cdot \alpha, \mathcal{D}) \end{aligned}$$

2. Backward Proof (from Knowledge PiNi to Hyperproperty PiNi)

Given that for an arbitrary program c where $c \in \text{KPiNi}_{\mathcal{D}}$, show that $\text{Behav}(c) \in \text{HPiNi}_{\mathcal{D}}$.

Lemma 4.1: Given $c \in \text{KPiNi}_{\mathcal{D}}$ and that c is deterministic, for any $m_1, m_2 \in \text{Store}$, $p_1, p_2 \in \mathbb{T}_{\text{fin}}$, $\alpha_1, \alpha_2 \in \mathbb{E}$ where $\neg \text{Sil}_{\mathcal{D}}(\alpha_1)$, $\neg \text{Sil}_{\mathcal{D}}(\alpha_2)$, $m_1 \approx_{\mathcal{D}} m_2$, $p_1 \approx_{\mathcal{D}} p_2$, $\langle c, m_1 \rangle \rightarrow_{p_1 \cdot \alpha_1}^*$ and $\langle c, m_2 \rangle \rightarrow_{p_2 \cdot \alpha_2}^*$: it follows that

$$\exists p \in \mathbb{T}_{\text{fin}}, \alpha \in \mathbb{E}. \neg \text{Sil}_{\mathcal{D}}(\alpha) \wedge \langle c, m_1 \rangle \rightarrow_p^* \wedge p \approx_{\mathcal{D}} p_2 \cdot \alpha \implies p_1 \cdot \alpha_1 \approx_{\mathcal{D}} p_2 \cdot \alpha_2$$

Proof: Assuming the premises:

- (a) $m_1 \approx_{\mathcal{D}} m_2$
- (b) $\langle c, m_1 \rangle \rightarrow_{p_1}^* \wedge \langle c, m_2 \rangle \rightarrow_{p_2 \cdot \alpha_2}^*$
- (c) $\exists p, \alpha. \neg \text{Sil}_{\mathcal{D}}(\alpha) \wedge \langle c, m_1 \rangle \rightarrow_p^* \wedge p \approx_{\mathcal{D}} p_2 \cdot \alpha$

it follows that $m_1 \in k_{\rightarrow}(c, m_2, p_2, \mathcal{D})$ directly from (D.10).

Since $c \in \text{KPiNi}_{\mathcal{D}}$ means $k_{\rightarrow}(c, m_2, p_2, \mathcal{D}) = k(c, m_2, p_2 \cdot \alpha_2, \mathcal{D})$, it means that $m_1 \in k(c, m_2, p_2 \cdot \alpha_2, \mathcal{D})$ as well.

From the definition of attacker knowledge, there exists some p'_1 such that $\langle c, m_1 \rangle \rightarrow_{p'_1}^*$ and $p'_1 \approx_{\mathcal{D}} p_2 \cdot \alpha_2$.

Since from premises $p_1 \approx_{\mathcal{D}} p_2$, it follows that $p_1 \approx_{\mathcal{D}} p_2 < p_2 \cdot \alpha_2 \approx_{\mathcal{D}} p'_1 \implies p_1 <_{\mathcal{D}} p'_1$.

As a consequence of determinism (NF1), it follows that $p_1 < p'_1$ (otherwise a contradiction arises from p'_1 both being shorter than p_1 and not being a prefix of p_1), and also that $p_1 \cdot \alpha_1 \leq p'_1$. From this, we can derive the desired result:

$$\begin{aligned} p_1 \cdot \alpha_1 &\leq_{\mathcal{D}} p'_1 \approx_{\mathcal{D}} p_2 \cdot \alpha_2 \\ p_1 \cdot \alpha_1 &\leq_{\mathcal{D}} p_2 \cdot \alpha_2 \\ p_2 \cdot \alpha_1 &\leq_{\mathcal{D}} p_2 \cdot \alpha_2 \\ \alpha_1 &= \alpha_2 \\ \implies p_1 \cdot \alpha_1 &\approx_{\mathcal{D}} p_2 \cdot \alpha_2 \end{aligned}$$

Lemma 4.2: Given $c \in \text{KPiNi}_{\mathcal{D}}$ and that c is deterministic, for any $m_1, m_2 \in \text{Store}$ and $p_1, p_2 \in \mathbb{T}_{\text{fin}}$: if $m_1 \approx_{\mathcal{D}} m_2$, $\langle c, m_1 \rangle \rightarrow_{p_1}^*$, and $\langle c, m_2 \rangle \rightarrow_{p_2}^*$ then

$$\text{len}(\lfloor p_1 \rfloor_{\mathcal{D}}) \leq \text{len}(\lfloor p_2 \rfloor_{\mathcal{D}}) \implies \lfloor p_1 \rfloor_{\mathcal{D}} \leq \lfloor p_2 \rfloor_{\mathcal{D}}$$

Proof: By induction on $\text{len}(\lfloor p_1 \rfloor_{\mathcal{D}})$:

Base Case: $\text{len}(\lfloor p_1 \rfloor_{\mathcal{D}}) = 0 \leq \text{len}(\lfloor p_2 \rfloor_{\mathcal{D}})$

if $\text{len}(\lfloor p_1 \rfloor_{\mathcal{D}}) = 0$, then $\lfloor p_1 \rfloor_{\mathcal{D}} = \epsilon$, for which $\epsilon \leq \lfloor p_2 \rfloor_{\mathcal{D}}$ holds trivially.

inductive Case: $\text{len}(\lfloor p_1 \rfloor_{\mathcal{D}}) = n + 1 \leq \text{len}(\lfloor p_2 \rfloor_{\mathcal{D}})$

As the inductive hypothesis, assume for any $\lfloor p'_1 \rfloor_{\mathcal{D}}$ with length n :

$$\text{len}(\lfloor p'_1 \rfloor_{\mathcal{D}}) \leq \text{len}(\lfloor p_2 \rfloor_{\mathcal{D}}) \implies \lfloor p'_1 \rfloor_{\mathcal{D}} \leq \lfloor p_2 \rfloor_{\mathcal{D}}$$

Since $\text{len}(\lfloor p_1 \rfloor_{\mathcal{D}}) = n + 1$, it follows that there exists $p_{1\mathcal{D}}, \alpha_1$ where $\neg \text{Sil}_{\mathcal{D}}(\alpha_1) \wedge \lfloor p_1 \rfloor_{\mathcal{D}} = p_{1\mathcal{D}} \cdot \alpha_1$.

Applying the inductive hypothesis shows that $p_{1\mathcal{D}} \leq \lfloor p_2 \rfloor_{\mathcal{D}}$. Combining this with the fact that $\text{len}(p_{1\mathcal{D}}) = n < \text{len}(\lfloor p_2 \rfloor_{\mathcal{D}})$ implies that there exists $p_{2\mathcal{D}}, \alpha_2$ where $p_{2\mathcal{D}} \cdot \alpha_2 \leq \lfloor p_2 \rfloor_{\mathcal{D}}$ and $p_{1\mathcal{D}} = p_{2\mathcal{D}}$.

Since $c \in \text{KPiNi}_{\mathcal{D}}$, c is deterministic, $m_1 \approx_{\mathcal{D}} m_2$, $\langle c, m_1 \rangle \rightarrow_{p_1}^*$, $\neg \text{Sil}_{\mathcal{D}}(\alpha_1)$, and $p_{1\mathcal{D}} \cdot \alpha_1 = p_{2\mathcal{D}} \cdot \alpha_1$, it follows from lemma 4.1 that

$$\begin{aligned} \lfloor p_1 \rfloor_{\mathcal{D}} &= p_{1\mathcal{D}} \cdot \alpha_1 = p_{2\mathcal{D}} \cdot \alpha_2 \leq \lfloor p_2 \rfloor_{\mathcal{D}} \\ \implies \lfloor p_1 \rfloor_{\mathcal{D}} &\leq \lfloor p_2 \rfloor_{\mathcal{D}} \\ \implies p_1 &\leq_{\mathcal{D}} p_2 \end{aligned}$$

The claim that $\text{Behav}(c) \in \text{HPiNi}_{\mathcal{D}}$ then follows. Let $(m_1, st_1), (m_2, st_2) \in \text{Behav}(c)$ be arbitrarily chosen trace productions s.t. $m_1 \leq m_2$. Then, arbitrarily pick $p_1, p_2 \in \mathbb{T}_{\text{fin}}$ s.t. $p_1 \leq st_1, p_2 \leq st_2$ and consider the comparison of $\text{len}(\lfloor p_1 \rfloor_{\mathcal{D}})$ and $\text{len}(\lfloor p_2 \rfloor_{\mathcal{D}})$.

Without loss of generality, assume $\text{len}(\lfloor p_1 \rfloor_{\mathcal{D}}) \leq \text{len}(\lfloor p_2 \rfloor_{\mathcal{D}})$. From lemma 4.2, it then follows that $\lfloor p_1 \rfloor_{\mathcal{D}} \leq \lfloor p_2 \rfloor_{\mathcal{D}} \implies p_1 \leq_{\mathcal{D}} p_2$ (from (NF.3)), as desired.

Lfp Proofs (LFP)

1. **Forward Proof (from Hyperproperty Lfp to Knowledge Lfp)** Given that for an arbitrary program c where $\text{Behav}(c) \in \text{HLfp}_{\mathcal{D}}$, show that $c \in \text{KLfp}_{\mathcal{D}}$.

Lemma 5.1: Given c s.t. $\text{Behav}(c) \in \text{HLfp}_{\mathcal{D}}$, for any $p \in \mathbb{T}_{\text{fin}}, m \in \text{Store}, \alpha \in \mathbb{E}$ s.t. $\neg \text{Sil}(\alpha)$ and $\langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$: it follows that for any $m' \approx_{\mathcal{D}} m \in \text{Store}$:

$$(\exists p'. p' \approx_{\mathcal{D}} p \wedge \langle c, m' \rangle \rightarrow_{p'}^*) \implies (\exists p^*, \alpha'. p^* \approx_{\mathcal{D}} p \cdot \alpha' \wedge \neg \text{Sil}(\alpha') \wedge \langle c, m' \rangle \rightarrow_{p^*}^*)$$

Proof: let $p \in \mathbb{T}_{\text{fin}}, m \in \text{Store}, \alpha \in \mathbb{E}$ s.t. $\neg \text{Sil}(\alpha), \langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$. Then, also let $p' \in \mathbb{T}_{\text{fin}}$ and $m' \approx_{\mathcal{D}} m \in \text{Store}$ s.t. $p' \approx_{\mathcal{D}} p$ and $\langle c, m' \rangle \rightarrow_{p'}^*$.

From (NF.2), finite traces $p \cdot \alpha$ and p' can be maximized: into some $st_1, st_2 \in \mathbb{T}$ where $p \cdot \alpha \leq st_2$ and $p' \leq st_1$

Since $p' \approx_{\mathcal{D}} p$ and $\neg \text{Sil}(\alpha)$, it follows that $p' <_{\mathcal{D}} p \cdot \alpha$. Applying defn. HLfp (D.13) then shows $\text{Prog}_{\mathcal{D}}(p', st_1)$, or $\exists p'' \leq st_1. p' <_{\mathcal{D}} p''$.

- Since $p' <_{\mathcal{D}} p'', \exists \alpha' \in \mathbb{E}$ where $\neg \text{Sil}(\alpha')$ s.t. $p' \cdot \alpha' \leq_{\mathcal{D}} p''$.
Then, from the definition of low prefix (D.9), it follows that $\exists p^* \leq p''$ for which $p^* \approx_{\mathcal{D}} p' \cdot \alpha' \approx_{\mathcal{D}} p \cdot \alpha'$
- since $p^* \leq p'' \leq st_1$, it follows from the definition of trace production (D.4) that $\langle c, m \rangle \rightarrow_{p^*}^*$

Therefore using p^* and α' as witnesses yields the desired result.

The claim that $c \in \text{KLfp}_{\mathcal{D}}$ then follows immediately from application of Lemma 5.1 as for any $p \in \mathbb{T}_{\text{fin}}, m \in \text{Store}$, and $\alpha \in \mathbb{E}$ where $\neg \text{Sil}(\alpha)$ and $\langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$:

$$\begin{aligned} k(c, m, p, \mathcal{D}) &= \{m' \mid m' \approx_{\mathcal{D}} m \wedge (\exists p' \in \mathbb{T}_{\text{fin}}. \langle c, m' \rangle \rightarrow_{p'}^* \wedge p \approx_{\mathcal{D}} p')\} \\ &= \{m' \mid m' \approx_{\mathcal{D}} m \wedge (\exists p^* \in \mathbb{T}_{\text{fin}}, \alpha' \in \mathbb{E}. \langle c, m' \rangle \rightarrow_{p^*}^* \wedge \neg \text{Sil}(\alpha') \wedge p^* \approx_{\mathcal{D}} p \cdot \alpha')\} \\ &= k_{\rightarrow}(c, m, p, \mathcal{D}) \end{aligned}$$

2. Backward Proof (from Knowledge Lfp to Hyperproperty Lfp)

Lemma 6.1: Given c s.t. $c \in \text{KLfp}_{\mathcal{D}}$, for all $p \in \mathbb{T}_{\text{fin}}, \alpha \in \mathbb{E}, m \in \text{Store}$ where $\neg \text{Sil}(\alpha)$ and $\langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$: it follows that for any $m' \approx_{\mathcal{D}} m \in \text{Store}$:

$$(\exists p'. p' \approx_{\mathcal{D}} p \wedge \langle c, m' \rangle \rightarrow_{p'}^*) \implies (\exists p^*, \alpha'. p^* \approx_{\mathcal{D}} p \cdot \alpha' \wedge \neg \text{Sil}(\alpha') \wedge \langle c, m' \rangle \rightarrow_{p^*}^*)$$

Proof: consider m, p s.t. $\langle c, m \rangle \rightarrow_{p \cdot \alpha}^*$, as well as m', p' where $m \approx_{\mathcal{D}} m', p \approx_{\mathcal{D}} p'$, and $\langle c, m' \rangle \rightarrow_{p'}^*$. From the attacker knowledge definition (D.6): $m' \in k(c, m, p, \mathcal{D})$

Then from $c \in \text{KLfp}$, it follows that $m' \in k_{\rightarrow}(c, m, p, \mathcal{D})$ as well.

From the progress knowledge definition (D.10), it follows that $\exists p^*, \alpha'. p^* \approx_{\mathcal{D}} p \cdot \alpha, \neg \text{Sil}(\alpha')$, and $\langle c, m' \rangle \rightarrow_{p^*}^*$, which is the desired result.

Let $(m_1, st_1), (m_2, st_2) \in \text{Behav}(c)$ where $m_1 \approx_{\mathcal{D}} m_2$.

Let $p_1 \in \mathbb{T}_{\text{fin}} \leq st_1$, from (D.4), it follows that $\langle c, m_1 \rangle \rightarrow_{p_1}^*$.

Let $p_2 \in \mathbb{T}_{\text{fin}} \leq st_2$ where $p_1 <_{\mathcal{D}} p_2$, from (D.4), it follows that $\langle c, m_2 \rangle \rightarrow_{p_2}^*$.

Since $p_1 \leq p_2$, it follows from the low prefix defn. (D.9) that $\exists p'_2 < p_2. p_1 \approx_{\mathcal{D}} p'_2$ and (D.4) that $\langle p, m_2 \rangle \rightarrow_{p'_2}^*$.

Applying Lemma 6.1 then shows $\exists p^*, \alpha'$ where $p^* \approx_{\mathcal{D}} p'_2 \cdot \alpha, \neg \text{Sil}(\alpha)$, and $\langle c, m_1 \rangle \rightarrow_{p^*}^*$. From there, maximalization (NF.2) and determinism show that $p^* \leq st_1$.

Then $p^* \approx_{\mathcal{D}} p'_2 \cdot \alpha \approx_{\mathcal{D}} p_1 \cdot \alpha$, and as $\neg \text{Sil}(\alpha): p^* >_{\mathcal{D}} p_1$, combining with the previous result shows that p^* can serve as the witness for $\text{Prog}_{\mathcal{D}}(p_1, t_1)$.

Thus, as both $p_1 \leq st_1$ and $p_2 \leq st_2$, as well as both (m_1, st_1) and $(m_2, st_2) \in \text{Behav}(c)$ are arbitrary, it follows that $\text{Behav}(c) \in \text{HLfp}_{\mathcal{D}}$, as desired.